Laughney 2020 LUAD — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

θ derived by connectivity (θ=5 fragments an 11% blob; θ=10 = one component). One patient (RU680, 125 cells) stays a same-patient clump at every θ (kNN-purity 16× vs ≤7× for the rest) — an unintegrable island, removed. Remaining 7/13 samples (2,734 cells) share one continuous manifold, max purity 6.6×, iLISI 1.00→3.24 — mixed & continuous.

Gate 2 — module geography → GO (non-branching)

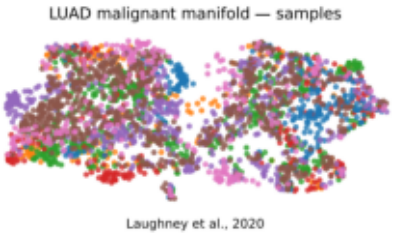
Single developmental axis driven by posterior-somitogenesis (pSom): pSom high at the origin declines along pseudotime (ρ=−0.87) opposed to a terminal neural-crest end that rises (ρ=+0.26; decile pt=0.68); Crest anti-correlated in space (Crest↔Gast −0.72, Crest↔pSom −0.45); Gast weak (ρ=+0.26, mid-axis) and ≈independent of pSom (r=−0.11). Joint pSom&Crest-high depleted (0.36) — inverse gradient, not a hub.

Gate 3 — cross-repression → NA

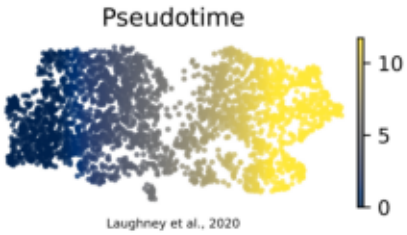
Single monotone axis — no bifurcation (curve: 2 tips, 0 forks) and no Mixed-type hub, so there is no branch pair to test for Gast↔pSom cross-repression.

STEP FIGURES

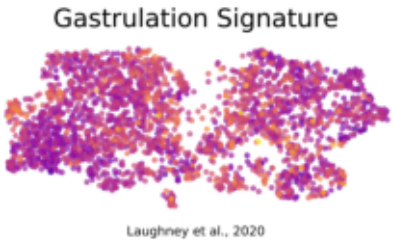
Manifold — samples



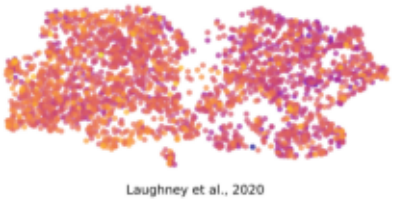
Pseudotime



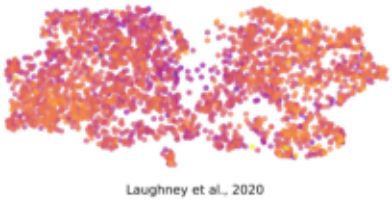
Gastrulation



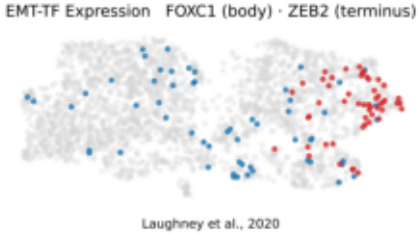
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Metastatic LUAD (Laughney et al., 2020), 10x — 13 patients / 40,505 cells (master scope); malignant only: 3,128 cells; ≥100-cell floor → 2,859 (8 samples), minus one unintegrable patient-private island (RU680, 125) → 2,734 cells / 7 patients on the trajectory (no scoping).

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Laughney's LUAD malignant manifold is a single monotone posterior-somitogenesis → terminal-neural-crest gradient (pSom ρ=−0.87, Crest ρ=+0.26); it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test (Linear-pSom/Crest class, shared with Qian / Kürten / Steele). Two decisive derived calls: integration strength θ=10 by connectivity, and removal of RU680 — a patient-private island that stays 16× kNN-pure at every θ, which the connectivity test alone had let pass. No cell-cycle annotation / flat proliferation, so no cycling lobe is scoped (contrast Qian). EMT terminus-loaded: expressed EMT-TFs lean late, ZEB2 marks the neural-crest terminus, no root-end factor (contrast Puram's SNAI1/ZEB1 pair).

Parameter choices

Cells / patients	2,734 / 7 (≥100 floor, −1 island)	**Integration**	Harmony θ=10 (connectivity); PCA50
Island removed	RU680 (125 cells; 16× pure, θ-inv.)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	pSom-high, Crest-low tip	**EMT-TFs**	terminus-loaded (ZEB2); no root factor